B.TECH CAPSTONE · COMPUTER SCIENCE AND ENGINEERING · BATCH MP-A05

Smart Ticket Understanding Engine

DRONA — intent classification, sentiment, SLA priority and routing for banking support tickets, offline on CPU

A transformer was not needed. Fine-tuned DistilBERT does edge out the cascade on accuracy — by 0.31 points. McNemar's test on the tickets where the two disagree returns p = 0.6061. That is not a result, it is noise. So we shipped the cascade: same accuracy, 3.2× faster, 168× smaller, and it carries a finite-sample guarantee on the tickets it auto-answers.

What we want to show: what we measured, what it cost, what we got wrong, and what we would do next. Every number in this deck is read from a JSON artifact at build time — none is typed in.

WHAT WE SHIPPED

94.60%
accuracy
macro-F1 94.43%

98.24%
on auto-answered
at 0.7635 confidence

33.3 ms
per ticket
in-process, CPU, no network

1.6 MB
total model size
vs 268 MB fine-tuned

TEAM MEMBERS

Four of us, one guide, and an industry mentor who read the draft.

B.Tech CSE, batch MP-A05 — Marri Laxman Reddy Institute of Technology and Management, under JNTUH.

#	REG. NO.	NAME	WHERE THE WORK SITS
1	237Y1A0540	**NAYAKAM SATYAJEET**	Pipeline, retrieval + fusion, calibration and the conformal gate
2	237Y1A0570	**JAKKU HARIKA**	DistilBERT fine-tune, the comparison, and the McNemar analysis
3	237Y1A0574	**SHERLA LOHITH KUMAR**	SLA priority, routing, sentiment and amount extraction
4	237Y1A05A5	**MAMIDALA VARUN**	FastAPI service, Streamlit console, drift monitor, evaluation

PROJECT GUIDE

Dr. PARVEZ

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INDUSTRY MENTOR

A practising industry expert from a HYSEA member company, who reviewed the capstone and gave feedback on the triage pipeline and its evaluation.

Submitted to Jawaharlal Nehru Technological University Hyderabad in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering.

PROBLEM STATEMENT

Nobody can staff a queue per intent. The label space is not the problem space.

Banking77: 13,083 real banking queries, 77 fine-grained intents, written by actual customers.

1 77 intents, ~8 real queues

77 intents collapse to 11 business categories. Routing to the fine label is not an option — no support organisation staffs it.

2 18.7:1 class imbalance

The largest category is 3,251 tickets; the smallest has 11. Accuracy alone is meaningless: predicting only the majority class already scores 24.8%.

3 High-consequence errors

A missed fraud ticket is a regulatory incident. A wrong priority misroutes a customer who is already angry. Both are expensive, and neither shows up in F1.

4 No safe way to abstain

The transformer tops accuracy at 94.91% but macro-F1 is only 94.07% — it wins overall and loses on the rare classes that matter.

AND THE PROBLEM WITH THE QUESTION

The brief asked us to beat *accuracy*. We took that as the starting question and found it was the wrong one. A system that is right 94.9% of the time and confidently wrong on the rare fraud categories is worse to deploy than one that is right 94.6% of the time and refuses to guess on the rest. **Everything we built after that is about making the refusal safe.**

OBJECTIVES

Four of the seven objectives in the synopsis are met outright. Two are partial, and one we deliberately did not do.

Quoted in intent from the synopsis. The status column is our own assessment against what we measured — not the synopsis's own claim.

#	OBJECTIVE (FROM THE SYNOPSIS)	STATUS	WHAT WE MEASURED
1	Classify tickets into enterprise business categories	MET	11 coarse categories from 77 intents; 94.60% accuracy, 94.43% macro-F1 on held-out test
2	Analyse sentiment and urgency for every ticket	MET	Sentiment lexicon plus category-priority rules and amount extraction; contributes to P0–P3
3	Assign an SLA-based priority P0–P3 and route to a queue	MET	Four priorities, per-category SLA targets and queues; P0 recall 95.49% on test
4	Compare TF-IDF, Logistic Regression, BiLSTM and DistilBERT	PARTIAL	Four of four arms run and compared on one split with a paired significance test — BiLSTM never built
5	Optimise DistilBERT with ONNX INT8 quantisation	NOT DONE	Declined deliberately: the cascade meets the latency and size goal with no transformer at all
6	Human-in-the-loop GenAI draft responses for agents	MET, NOT ADOPTED	Agent drafts with fact precision 1.000 and 0 invented ids — but letting it decide lowered accuracy, so it ships advisory-only
7	Deploy as containerised microservices with tracking and drift monitoring	PARTIAL	FastAPI + Streamlit + drift monitor, 8 experiments tracked end to end — no containers

Objective 5 was to quantise DistilBERT with ONNX INT8 to cut latency and size. We did not do it, because **we shipped a system that is faster and smaller without it** — 33.3 ms and 1.6 MB against 106 ms and 268 MB. The honest reading is that we met the goal and declined the method.

LITERATURE SURVEY

Every component we needed already existed. Assembled, calibrated and measured together, they did not.

The nine works in the synopsis, plus the four we had to add because the work needed them.

AUTHOR & YEAR	TECHNIQUE	LIMITATION WE INHERITED
Paramesh & Shreedhara, 2019	ML classification of IT service-desk tickets	Classical features only — no sentiment, SLA or routing
Zangari et al., 2023	Survey of ticket automation and multi-level classification	Survey; no deployable low-latency pipeline
Casanueva et al., 2020	Dual sentence encoders; introduces Banking77	Intent detection only — no priority or routing
Devlin et al., 2019	BERT, bidirectional pretraining and fine-tuning	Large; high latency and memory cost on CPU
Sanh et al., 2019	DistilBERT — ~40% smaller, ~60% faster than BERT	Needs further optimisation for CPU serving
Zafrir et al., 2019	Q8BERT, 8-bit quantization-aware training	Requires retraining; evaluated on general benchmarks
Minaee et al., 2021	Review of CNN / RNN / attention / transformer classifiers	Survey; no deployment guidance
Lewis et al., 2020	Retrieval-augmented generation for knowledge-intensive NLP	Needs a knowledge index; hallucination risk
Zaharia et al., 2018	MLflow — tracking, packaging, model registry	Tracking only; no text-drift monitoring

AND THE FOUR WE ADDED

Tunstall et al., 2022 · SetFit, arXiv:2209.11055

8 examples per class beats full fine-tuning by 19.3 points — the sanity check on how little data a good head needs.

Angelopoulos & Bates, 2019 · conformal prediction, arXiv:1907.07509

Turns "auto-answer the confident 90%" into a finite-sample guarantee rather than a tuned threshold.

Northcutt et al., 2021 · confident learning / cleanlab

The only way to know whether the labels we trained on are wrong is to audit them.

Ying & Thomas, 2022 · Banking77 label noise, 2022.insights-1.19

~14% noise reported at the 77-intent level; ours found far less, and the gap is worth explaining.

THE GAP

Nine works, nine limitations — and every one of them stops exactly where the next one starts. Classifiers that do not route. Routers that do not prioritise. RAG that hallucinates. MLOps that watches metrics but not text. Not one of them ships a calibrated, abstaining, CPU-only triage loop.

A disagreement worth reporting. Ying & Thomas report ~14% label noise in Banking77; our confident-learning audit flags 101 of 9,158 training labels — 1.1%. Not the same claim: theirs is at the 77-intent level on the full dataset, ours at the coarse level on the train split only: coarse grouping absorbs most fine-intent boundary noise.

METHODOLOGY

Two sparse views, two cheap heads, one calibrated gate. Every number below is measured on a held-out split.

Train 9,158 / validation 1,962 / test 1,963, stratified 70/15/15 on the coarse label, seed 20261003. Nothing is tuned on test.

1 TWO VIEWS
char_wb 3–5 grams
word 1–2 grams
34,901 features

→

2 TWO HEADS
LinearSVC
kNN, k=15
trained on the same union

→

3 FUSION
weighted log-prob average, then temperature
α=0.6, T=0.75

→

4 CALIBRATE
vector scaling on the validation split
ECE 0.0170

→

5 GATE
conformal threshold at 90% target coverage
q=0.3032

→

6 ACT
auto-answer, or escalate + route + prioritise
0.7635 served

DATA

Source Banking77 (Casanueva et al., 2020), CC BY 4.0

Rows 13,083 tickets, 77 intents

Labels 11 coarse, 10 reachable

Empty class loan_emi — no examples in the dataset at all

Split stratified 70/15/15, seed 20261003

Leakage control α and T grid-searched on validation only

WHAT WE COMPARED

kNN retrieval no training, neighbours vote

LinearSVC 94.50% on the same features

Cascade the two fused, calibrated and gated

DistilBERT 2 epochs, 47 min CPU

BiLSTM in the brief, never built — see slide 4

Agent advisory reranking on escalated tickets only

WHAT WE HELD FIXED

One split every arm scored on the same 1,963 test tickets

No early stop chosen on validation, then frozen

Paired test McNemar, because the arms are not independent

Reporting accuracy AND macro-F1 AND ECE, never one alone

Reproducibility 8 experiments → 9 JSON artifacts

SYSTEM ARCHITECTURE

A retriever, a linear head, a fusion, and a gate. The transformer is not in this diagram.

One ticket in, one routed and prioritised ticket out. No network call, no model download, no API key at any point in the path.

INPUT

Raw ticket text
what the customer wrote

Two sparse views
char_wb 3–5 + word 1–2 → 34,901 dims

kNN neighbours
k=5 served; oracle ceiling 99.75% at k=15

DECIDE

LinearSVC head
one weight row per coarse class

kNN head
15 neighbours vote, distance-weighted

Fusion
α=0.6 mix, then T=0.75 temperature

Vector scaling
fitted on validation; ECE on test below

Conformal gate
α=0.1 split-conformal on the val split

ACT

Auto-answer
above 0.7635 — 98.24% accurate

Escalate
below threshold → a human decides

Priority
category impact floor + amount at risk → P0…P3

Route
SLA queue, from the category and the priority

Draft
agent writes, human sends — never the reverse

WHAT THIS DIAGRAM EXCLUDES

No transformer. No LLM in the decision path. The fine-tuned DistilBERT is a measured arm of the experiment — 94.91% accuracy, 106 ms, 268 MB, 47 min to train — and it lost on p = 0.6061. The generative agent is in the diagram as a drafter, never as a decider: given decision authority on the 129 escalated tickets it moved end-to-end accuracy 94.70% → 94.40%.

PROJECT DESIGN

Four decisions that shaped everything else. Each one is a trade we made on purpose.

Design is what you removed. Each row is a choice we could have gone the other way on, and what it cost.

DECISION	THE ALTERNATIVE	WHY WE CHOSE THIS	WHAT IT COST US
Sparse views, not embeddings	Sentence-transformers or fine-tuned embeddings	char_wb 3–5 grams catch the misspellings and the code-switching that actually appear in support tickets; a static vocabulary also means nothing to download at serving time.	No semantic generalisation to a phrasing we never saw.
A linear head, not a neural classifier	Fine-tuned DistilBERT as the primary classifier	It matched a 268 MB transformer to within 0.31 points, and a paired test put that margin at chance.	Linear models cannot represent word order — see the sentiment limitation.
Fusion, then calibration, then a gate	One classifier at one threshold	Fusing two decorrelated errors raised macro-F1, and calibration is what turns a probability into something an operator can act on.	Two more fitted stages between the text and the answer.
Conformal gate on the validation split	Pick a threshold on the test curve	A threshold tuned on test is a number that cannot survive contact with new data; split conformal gives a bound that holds on fresh exchangeable tickets.	Coverage is 90.32% rather than exactly 90%.
Agent drafts, humans decide	Let the agent overrule the classifier	On the 129 escalated tickets the agent's own accuracy was lower than the cascade it was asked to override.	None of the drafting speed-up reaches the customer.
Report ECE and macro-F1 next to accuracy	Report accuracy	Under 18.7:1 imbalance the two disagree, and a model can be well-ranked and badly calibrated at the same time.	More numbers to read before you can compare two systems.

And one non-decision we want on the record: 77 intents map to 11 categories, and loan_emi has no examples in the dataset. We left the class in the taxonomy rather than deleting it, because a routing table that quietly drops a category is worse than one that admits the gap.

EXPECTED OUTCOMES

We expected the transformer to win. It won by a margin that does not survive a significance test.

Four systems, one held-out split of 1,963 tickets, one test applied to the pair that disagreed.

SYSTEM	ACCURACY	MACRO-F1	ECE	LATENCY	SIZE	BUILD
kNN retrieval only	92.56%	91.75%	0.0785	27.4 ms	—	none
TF-IDF + LinearSVC	94.50%	94.31%	uncalibrated	<0.01 s	—	2.8 s fit
Cascade — retrieval + linear	**94.60%**	94.43%	0.0170	33.3 ms	1.6 MB	not timed
DistilBERT fine-tuned	94.91%	94.07%	0.0252	105.5 ms	268 MB	47 min

McNemar, ON THE 100 TICKETS WHERE THEY DISAGREE

	Cascade right	DistilBERT right	Total
Cascade right	1,813	**50**	1,863
DistilBERT right	**44**	56	100
Total	1,857	106	1,963

DistilBERT wins 50 tickets and loses 44. $\chi^2 = 0.266$, $p = 0.6061$, exact paired test. That is the entire basis for preferring a transformer.

And on the rare classes the two arms tie exactly: tail recall 90.38% and 90.38%. The transformer is not the intents that are hard to predict -- it is the same model on that slice, for 3.2x the latency.

SO WE SHIPPED THIS INSTEAD

98.24% accuracy on the tickets it auto-answers

96.84% accuracy where the two heads agree

90.32% coverage actually delivered at α = 0.10

100.00% accuracy on those accepted tickets

Expected outcome, restated honestly: not "higher accuracy", but a system whose accuracy we can promise, on the subset it acts on, at a coverage rate we chose deliberately. The guarantee held at every α we tested. It cost 3.2× less latency and 168× less disk than the arm we rejected.

FUTURE PLAN

What we would do next, and what we would not.

WE WOULD FIX FIRST

Sentiment is the weakest link

It is a lexicon, and lexicons invert on negation ("not working" scores positive). A small supervised head on the same features would fix it.

BiLSTM was never built

It sits between a bag-of-words and DistilBERT. The comparison in the synopsis is incomplete and we should say so plainly.

The label space is still wrong

Real desks do not have 11 categories. Mapping the taxonomy to an actual bank's queue structure needs their input, not ours.

WE WOULD ADD NEXT

A real abstention interface

The conformal gate works; nothing yet tells an agent *why* a ticket was escalated. Retrieval evidence is already computed — we just do not show it.

Drift monitoring on text, not just labels

The monitor watches category mix. It should watch embedding drift too; the current reference is drawn from train, so it can only fire on mix.

Per-class thresholds

Conformal gives one global q. P0 is worth a stricter gate than P3, and the sweep says the margin matters more than the score.

WHAT WE WOULD NOT DO

Adopt the transformer for the 0.31 points

$p = 0.6061$. To make that margin real we would need roughly ten times the test set — and then it still has to win on ECE.

Give the agent decision authority

94.70% → 94.40% on the 129 escalated tickets. Drafting is where it adds value.

Auto-answer on confidence alone

That is the number 0.80 that means 95% accuracy in one version and 70% in the next. Coverage is the promise; the threshold is an implementation detail.

The one-line version: the useful part of this project was not reaching 94.60%. It was building a system that knows the difference between 94.91% and 94.60% — and refusing to act on the difference.